A PROJECT REPORT

ON

NUTRI LENS AI

A report submitted in partial fulfilment of the requirements for the award of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (AI&ML)

SUBMITTED BY

UNDER THE ESTEEMED GUIDANCE OF

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (AI&ML)



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

(AI&ML)

St. Peter's Engineering College (UGC Autonomous)

Approved by AICTE, New Delhi, Accredited by NBA and NAAC with 'A' Grade,

Affiliated to JNTU, Hyderabad, Telangana.

2025-2026

DEPARTMENTOF COMPUTER SCIENCE & ENGINEERING (AI&ML)

CERTIFICATE

This is to certify that project entitled**"EDUCATION BEYOND BORDERS"** is carried out by TANDRA BHARGAV (24BK1A66CP), PADAMPALLY AKHIL (24BK1A66CA), RANGU LOKESH (24BK1A66CJ), in partial fulfilment for the award of the degree of **Bachelor of Technology in COMPUTER SCIENCE & ENGINEERING (AI&ML)** is a record of bonafide work done by them under my supervision during the academic year "2025– 2026".

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We respect and thank our Administrative Director **Mr. T. Anuraag Reddy**, Academic Director **Mrs. T. Saroja Reddy** and secretary **Sri. T. V. Reddy**, for providing us an opportunity to do the project work at **St. PETER'S ENGINEERING COLLEGE** and we are extremely thankful to them for providing such a nice support and guidance which made us to complete the project.

We also acknowledge with a deep sense of reverence, our gratitude towards our parents, who have always supported us morally as well as economically. We also express gratitude to all our friends who have directly or indirectly helped us to complete this project work.

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INSTITUTE VISION

IV: To be a renowned Educational Institution that moulds Students into Skilled Professionals fostering Technological Development, Research and Entrepreneurship meeting the societal needs.

INSTITUTE MISSION

IM1: Making students knowledgeable in the field of core and applied areas of Engineering to innovatetechnological solutions to the problems in the society.

IM2: Training the Students to impart the skills in cutting edge technologies, with the help of relevant stakeholders.

IM3: Fostering a conducive ambience that inculcates a research attitude, identifying promising fields forentrepreneurship with ethical, moral and social responsibilities.

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To improve fundamentals and make it easier to address the ever expanding needs of society by producing the best leaders in artificial intelligence and machine learning via excellence in research and education.

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- **DM 2 :** Impart rigorous training to generate knowledge through the state-of-the-art concepts and technologies in Artificial Intelligence and Machine Learning.
- **DM 3 :** To enhance research in emerging areas by collaborating with industries and institutions at the national and international levels.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- **PEO 1 :** Work effectively in inter-disciplinary field with the knowledge of Artificial Intelligence and Machine Learning to develop solutions to the real-world problems.
- **PEO 2 :** To communicate and work effectively on team based engineering projects and will practice the ethics of their profession consistent with a sense of social responsibility.
- **PEO 3 :** Excel as socially committed engineers or entrepreneurs with high ethical and moral values.

PROGRAM OUTCOMES (POs)

PO1- Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and provide solutions in the engineering specialization of artificial intelligence and machine learning.

PO2- Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3- Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, and environmental considerations.

PO4- Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5- Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6- The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

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PO8- Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9- Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10- Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective Presentations, and give and receive clear instructions.

PO11- Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary Environments.

PO12- Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

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PSO1: To apply knowledge of Data Science, Machine Learning, and AI in exploring, adopting, and integrating emerging technologies for practical, research, and co-curricular projects.

PSO2: The ability to design and develop software and data-driven projects using standard practices and modern tools.

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (AI&ML)

DECLARATION

We declare that a Major Project entitled **"EDUCATION BEYOND BORDERS"** is an original work submitted by the following group members who have actively contributed and submitted in partial fulfilment for the award of degree in **"Bachelor of Technology in Computer Science and Engineering (AI&ML)"**, at **St. Peter's Engineering College, Hyderabad,** and this project work has not been submitted by me to any other college or university for the award of any kind of degree.

Group No:05

Program: B.Tech

Branch: CSE(AI&ML)

Major Project Title: EDUCATION BEYOND BORDERS

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ABSTRACT

The Education Beyond Borders is a global initiative aimed at making quality learning accessible to all, regardless of nationality, geography, or socio-economic background. The project focuses on creating an interconnected educational ecosystem that leverages digital technology, cross-cultural collaboration, and open learning resources. By removing physical and economic barriers, the initiative seeks to empower learners in remote, underprivileged, and conflict-affected regions with opportunities equal to those available in developed areas. The project integrates online learning platforms, virtual classrooms, multilingual content, and interactive digital tools to support inclusive and flexible education. It encourages collaborative learning between students and teachers from different countries, enhancing cultural understanding and global citizenship. Through partnerships with educational institutions, NGOs, and technology providers, Education Beyond Borders promotes the sharing of resources, best practices, and innovative teaching methods. A key component of the project is its commitment to equity, ensuring that marginalized groups—such as girls, refugees, and rural learners—receive targeted support and accessible learning pathways.

TABLE OF CONTENTS

Ch. No.	Title of the Chapter	Page No.
Certificate		ii
Acknowledgement		iii
Declaration		viii
Abstract		ix
CHAPTER I - INTRODUCTION		
1.1	Introduction	1
1.2	Objective of the project	2
1.3	Organizational of the thesis	2
CHAPTER 2- LITERATURE SURVEY		
2.1	Literature survey	4
2.2	Existing works	5
2.3	Limitations of existing works	6
2.4	Advantages of proposed solution	6
CHAPTER 3 – ANALYSIS		
3.1	Problem statement	8
3.2	Software requirements	8
3.3	Hardware requirements	9
CHAPTER 4- DESIGN		
4.1	System architecture	10
4.2	Use case diagram	11
4.3	Activity diagram	13
4.4	Sequence diagram	14

Chapter 5- IMPLEMNTATION AND RESULTS

5.1 Data preparation and feature selection 16

5.2 Model implementation 16

5.3 Model evaluation metrics 17

5.4 Experimental results 18

5.5 Deployment and user interface 18

Chapter 6- TESTING AND VALIDATION

6.1 Training process 20

6.2 Testing process 21

6.3 Validation metrics 22

Chapter 7- CONCLUSIONS

7.1 Conclusion 23

7.2 Future Scope 23

REFERENCES 24

SOURCE CODE 25.

LIST OF TABLES

Table No.	Particulars	Page No.
3.2	Software requirements	8
3.3	Hardware requirements	9
6.3	Validation metrics	22

LIST OF FIGURES

Fig. No.	Particulars	Page No.
4.2	Use case diagram	12
4.3	Activity diagram	14
4.4	Sequence diagram	15

CHAPTER- I

INTRODUCTION

1.1 INTRODUCTION

Packaged foods are convenient, but their labels can be difficult to read quickly. Small print, curved wrappers, and unfamiliar ingredient names make it easy to miss an allergen or additive. For people managing allergies, diabetes, hypertension, or dietary restrictions, one overlooked term can matter. NutriLens AI was developed to make those labels easier to review at the point of purchase. This need is practical, not limited to specialists. A clear explanation helps shoppers check a product before they put it in the basket with confidence.

Many food apps depend on barcode records, which may not include local products or recently changed recipes. NutriLens AI reads the ingredient panel from a photograph, cleans the image, and uses optical character recognition (OCR) to capture the text. A parser separates ingredient names and maps familiar aliases to standard records. Shoppers can therefore review a package even when no matching product listing is available.

The application compares parsed ingredients with the shopper's saved allergies, health conditions, diet choices, and avoid list. It returns SAFE, CAUTION, or AVOID and identifies the ingredient and rule behind each warning. When the database contains a suitable option, the app can suggest an alternative. Shoppers can review unclear text and revisit scan history. NutriLens AI supports informed choices; it does not replace a clinician's advice.

1.2 PROJECT OBJECTIVES

The project sets out five practical objectives for shoppers:

· Capture food labels with a camera and keep each scan usable when lighting, glare, or package curves make photography difficult.

· Improve label images and extract ingredient text before checking the words against the shopper's saved personal health profile.

· Normalize ingredient names, common aliases, and additive codes so the same substance can be recognized across different printed label wording.

· Compare each parsed ingredient with allergies, health needs, diet choices, and avoid entries saved by the shopper.

· Return a clear SAFE, CAUTION, or AVOID status, explain each match, and suggest alternatives when suitable verified product records are available.

1.3 ORGANIZATION OF THE REPORT

This report follows the project through seven chapters.

Chapter 1 – Introduction

Chapter 1 introduces the food label problem, explains why ingredient names can be hard to assess, and presents the project's purpose, objectives, scope, and the risks that NutriLens AI helps shoppers review.

Chapter 2 – Literature Survey

Chapter 2 reviews food scanning apps, barcode databases, image recognition, and OCR research. It compares current tools and explains why direct label reading and profile-based guidance are useful additions. It connects the literature review with the practical problem addressed by the application.

Chapter 3 – System Analysis

Chapter 3 defines the problem and lists the software and hardware needed to run the application. It describes the main requirements, system users, and practical operating limits.

Chapter 4 – System Design

Chapter 4 presents the system architecture, database, user roles, and workflow diagrams. It follows a label image through OCR, ingredient parsing, profile checks, and the result shown to the shopper. The design chapter follows a label from image capture through OCR, parsing, profile checks, and the result for shoppers.

Chapter 5 – Implementation and Results

Chapter 5 describes the ingredient records, image-processing pipeline, parser, personalization rules, evaluation measures, and web interface used in the application.

Chapter 6 – Testing and Validation

Chapter 6 explains how the knowledge base and application were checked. It summarizes cases for OCR, ingredient matching, profile rules, scan results, and usability.

Chapter 7 – Conclusions and Future Scope

Chapter 7 summarizes the project and its limits, then outlines practical next steps such as broader language support and more capable on-device image recognition. The final chapter separates completed work from future ideas.

CHAPTER- II

LITERATURE SURVEY 2.1

LITERATURE SURVEY

Digital food tools have grown alongside smartphone cameras and online nutrition databases. Early nutrition apps mainly asked people to search for a product or enter its details by hand. Barcode scanning made that process quicker, but it still depends on a product record being available and current. Those tools are useful for familiar products, yet they may not describe a local brand or a recently updated recipe.

Research on package reading has also looked at optical character recognition applied to label photographs. Flexible packaging creates practical challenges: text may curve around a container, glare can hide letters, and decorative printing can confuse recognition. Image cleanup, contrast adjustment, and deskewing can make the text easier for OCR to read, although results still depend on photo quality and label layout.

Food databases and allergy tools provide useful reference information, but a standard score cannot reflect every shopper's needs. A product can suit one person and be unsuitable for another. NutriLens AI builds on these earlier ideas by reading the ingredient panel itself and checking recognized terms against a saved profile. Its result also shows which ingredient and rule led to a warning. Image quality and saved records affect scan results in practice.

2.2 EXISTING FOOD LABEL TOOLS

Open Food Facts is a collaborative product database with ingredient and nutrition details. It can be helpful when a product has already been submitted, but a barcode lookup may not find an unlisted or regional item. A barcode record helps when it is current. It supplies product details, but cannot show whether ingredients fit a personal health profile or dietary choices.

Yuka scans product barcodes and presents a general nutrition and additive score. That summary is easy to review, though it is not a substitute for checks based on a person's allergies or health profile. NutriLens AI takes a different path: it reads package text, resolves known aliases, and checks the resulting records against a saved profile.

Personalized allergy apps such as Fig focus on dietary restrictions and product matching. Their usefulness depends on the coverage and freshness of the product records behind the service.

Barcode services can save time when a reliable match exists. They may leave a shopper without an answer when a package is new, locally made, or missing from the catalogue. The two approaches can work together. A stored entry can verify product details, while a photograph lets a shopper inspect a label directly.

NutriLens AI complements those services by reading a photographed ingredient list. Its profile rules add personal context, while the database still supplies canonical names, categories, explanations, and alternatives. A barcode lookup is useful when the product is listed.

2.3 LIMITATIONS OF CURRENT TOOLS

Existing food tools are helpful, but several limits remain when shoppers need to check a package in the aisle.

- **Barcode lookup misses some products**
  Barcode coverage is uneven. A new recipe, a small local brand, or a product from another region may not have a usable database entry.
- **General scores are not personal**
  General nutrition grades apply the same scoring approach to everyone. They cannot fully account for a particular allergy, dietary choice, or health condition.
- **OCR can miss label details**
  Label photographs are not always clean. Glare, small print, curved surfaces, and nested ingredient groups can lead to OCR mistakes or missed terms.
- **Warnings need clear explanations**
  Some apps show a score without naming the ingredient that affected it. Shoppers may also receive few practical suggestions when an item does not fit their needs. These gaps matter when shoppers need an explanation, not just a grade. Clear reasons show which ingredient triggered the alert.

2.4 THE PROPOSED NUTRILENS APPROACH

NutriLens AI is designed to close these gaps with direct label reading and profile-based checks.

- **Direct label reading**
  A shopper can scan a package photograph without first finding its barcode record. This helps with local foods and products that have not yet appeared in a catalogue.

- **Ingredient alias matching**

  Image cleanup and OCR turn visible label text into words that the application can review. The result still depends on image quality, so unclear text remains visible for checking.

- **Profile-based safety checks**

  The parser maps common aliases and additive codes to canonical ingredient records. That makes it easier to recognize a familiar substance when a label uses a different name.

- **Clear reasons for warnings**

  The safety check uses the shopper's saved allergies, health needs, diet choices, and personal avoid list. The same food can therefore receive different guidance for different profiles.

- **Review when OCR is uncertain**

  Each warning names the ingredient that matched and explains the rule behind the result. This gives the shopper a reason to review rather than only a single score.

- **Scan history and alternatives**

  Unknown terms are treated cautiously and can be reviewed against the package. They are not presented as confirmed safe simply because no database match was found.

- **Personal avoid lists**

  When the knowledge base contains a relevant option, the application can suggest an alternative. Scan history also lets shoppers revisit earlier checks and explanations.

- **A modular application**

  The Flask application keeps image reading, parsing, database access, and profile checks in separate services. That structure makes individual parts easier to review and maintain.

CHAPTER- III

SYSTEM ANALYSIS

3.1 PROBLEM STATEMENT

Packaged food labels can be hard to assess during a quick shopping trip. Print may be small or wrapped around a container, and ingredient names may use technical terms, aliases, or additive codes. A shopper managing an allergy, diabetes, hypertension, or a dietary restriction may need to recognize those terms before choosing a product. Barcode apps help when a current product record exists, but do not cover every local or newly released item. The project addresses this gap by reading the printed ingredient panel from a camera image and making its contents easier to review.

NutriLens AI cleans the image, extracts text, normalizes ingredient names, and compares them with a personal profile. It must identify the ingredient and rule behind each warning, keep unknown terms marked for review, and handle unreadable photographs without guessing. The result is a decision aid for shoppers, not a diagnosis or a replacement for medical advice. A missing record needs careful review; no match is not evidence of safety for shoppers.

3.2 SOFTWARE REQUIREMENTS

NutriLens AI uses Python, Flask, OpenCV, Tesseract OCR, MySQL, and a browser interface to read labels and check ingredients against profiles.

Software	**Purpose / Description**
Operating system	Windows 10/11 or Linux
Programming Language	Python 3.11 runtime, 64-bit core
Development Environment	VS Code

Libraries Used	OpenCV, Tesseract OCR, PyMySQL
Web Application Framework	Flask, Jinja2, HTML, CSS, JavaScript
Ingredient Storage	MySQL relational database tables

3.3 HARDWARE REQUIREMENTS

The application runs on a standard computer or phone with a camera, enough memory for image processing, and storage for ingredient records and scan history.

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i3 or equivalent	Intel Core i5 or higher
RAM	4 GB	8 GB or higher
Storage	500 MB free space	1 GB or higher
Camera	Phone camera or uploaded image	Autofocus camera, good lighting
Supported Operating System	Windows 10	Windows 11 or Linux

CHAPTER- IV

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

NutriLens AI uses a modular four-part design. The browser, Flask application, processing services, and database each have a clear role in a label scan. This separation keeps interface and database work distinct during each scan.

1. Architecture overview

The main parts of the system are:

- Presentation: a responsive HTML, CSS, and JavaScript interface lets shoppers capture a label, upload a photograph, and review the result.
- Flask application: routes requests, manages the user session, and passes submitted images or text to the processing services.
- Image and ingredient services: OpenCV prepares the photograph, Tesseract extracts text, and the parser separates and normalizes ingredient names.
- Personalization rules: the engine compares ingredient records with allergies, health needs, dietary choices, and the shopper's avoid list.
- MySQL stores profile details, ingredient records, aliases, and scan history so the application can retrieve and update them.

4.2 USE CASE DIAGRAM

The use case diagram shows how a shopper and the application work together during a label scan. The diagram separates shopper actions from work handled by supporting services.

Actors:

1.1 Shopper

The shopper is the primary user.

The shopper is the primary user of the application.

1.2 OCR service

A shopper can create a profile with allergies, health conditions, dietary choices, and ingredients to avoid. Only profile fields are used in the comparison.

The OCR service reads ingredient text from a captured or uploaded package image.

1.3 Database administrator

Image cleanup improves contrast and corrects skew before text recognition begins.

The ingredient parser separates names and expands nested ingredient groups.

Use Cases:

1. Set a personal food profile

The database administrator maintains ingredient records, aliases, allergen groups, explanations, and alternatives.

2. Capture or upload a label

A shopper can update profile details at any time. These settings guide the safety checks but do not change the printed ingredient list. A scan can begin with a camera capture or an image stored on the device.

3. Review ingredients and alerts

The user captures a package label with a camera or selects an existing image. Manual text can be reviewed when a photograph is difficult to read. If extracted text is incomplete, the shopper can inspect the package and enter a correction.

4. View safety results

OCR extracts visible words. The parser removes formatting noise, separates ingredients, and checks common aliases and additive codes.

Scan history and explanations

The safety engine compares normalized ingredients with allergies, health conditions, diet choices, and the personal avoid list. Each warning stays connected to a recognized ingredient and its matching profile rule.

NutriLens AI presents SAFE, CAUTION, or AVOID with the ingredient match and a plain explanation of the result.

When the database contains a relevant option, the result can include an alternative. The shopper can review the package whenever a term is uncertain. Alternative suggestions depend on a relevant database record.

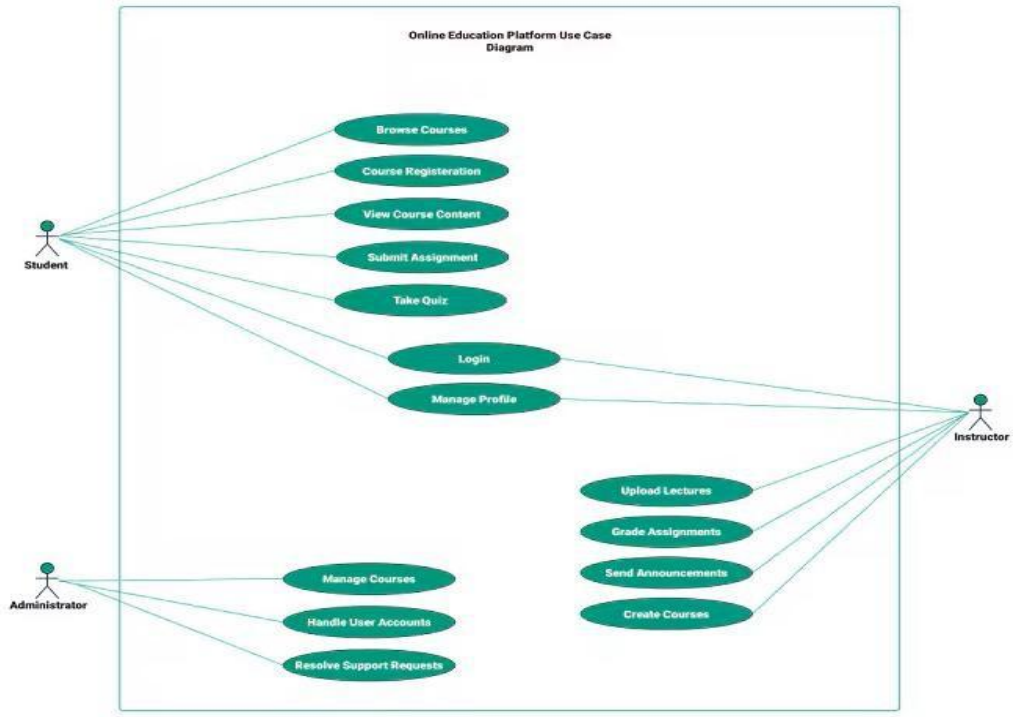


Figure 4.2: NutriLens AI Food Safety Use Cases

4.3 ACTIVITY DIAGRAM

The activity diagram follows a shopper from profile setup through label capture, OCR, ingredient checks, and a clear safety explanation.

Flow of Activities:

- **Set profile and allergies**
- **Capture or upload a label**
- **Review ingredient analysis**
- **Check safety warnings**
- **Check allergen and health rules**
- **Review scan history**
- **Read label with OCR**
- **Maintain ingredient records**
- **Review results and alternatives**

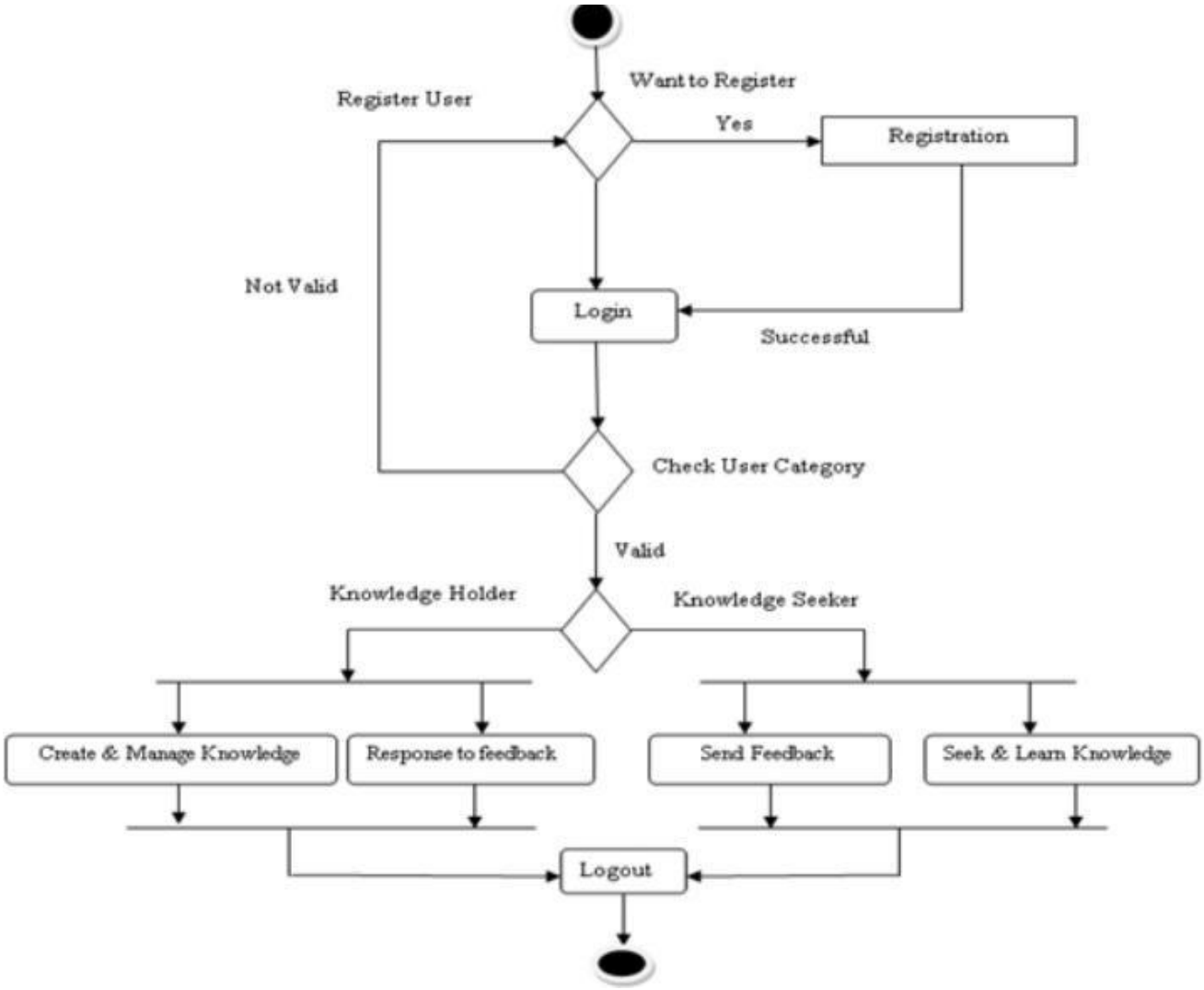


Figure 4.3: Activity Diagram /

4.4: Sequence Diagram

The sequence diagram follows a label image as it moves between the browser, Flask, OCR, parser, rules, and database.

Each service returns its result to the interface for review.

Main Services:

- **Shopper**
- **Web interface**
- **OCR and safety services**

- **Ingredient**

database / sequence of

actions:

1. Shopper → Interface: Captures or uploads a label.
2. Interface → OCR: Sends the image for text extraction.
3. Parser → Database: Resolves names and ingredient aliases.
4. Rules → Interface: Returns the status and explanation.
5. Interface → Shopper: Displays warnings and alternatives.

This flow connects the interface, image processing, ingredient checks, and the shopper's result while keeping each step traceable.

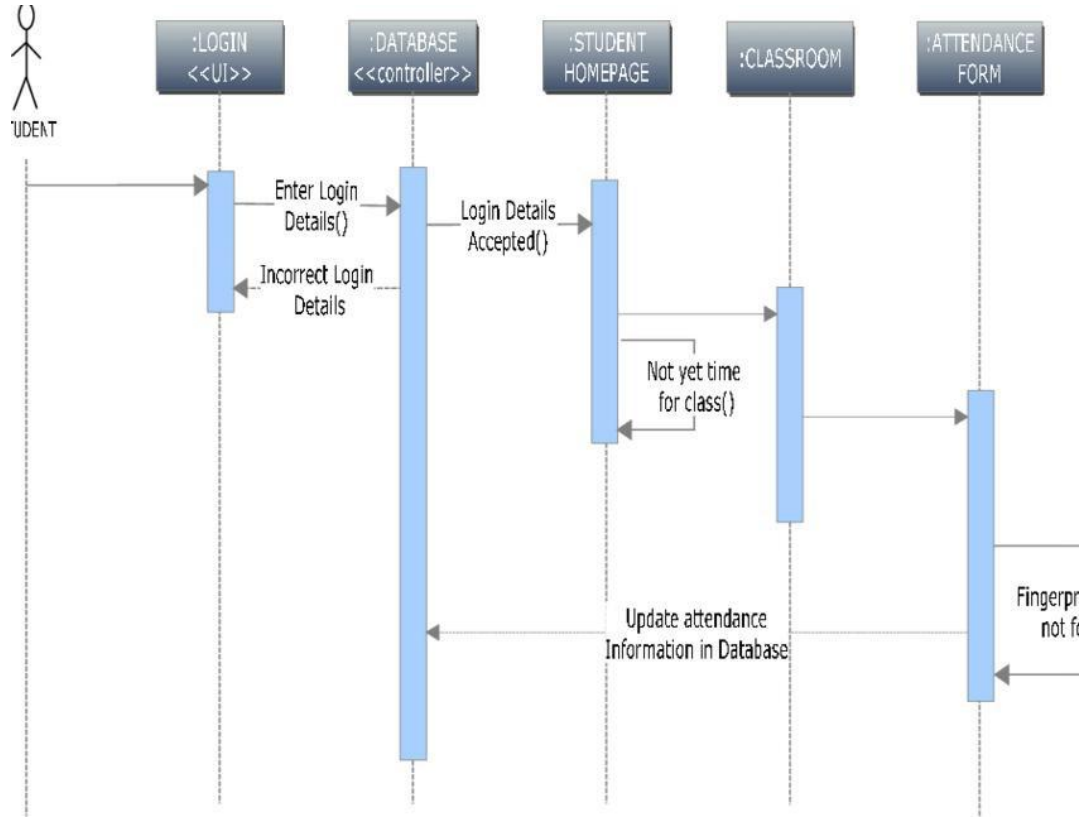


Figure 4.4: Label Scan Sequence

CHAPTER- V

IMPLEMENTATION AND RESULTS

5.1 INGREDIENT DATA AND KNOWLEDGE BASE

Ingredient data review

NutriLens AI uses an ingredient knowledge base rather than a trained classification model. Records store canonical names, aliases, additive codes, allergen and diet categories, explanations, and alternatives. The parser maps label synonyms to canonical records before profile rules run. The database also stores personal avoid entries and scan history. Reviewers check spelling, categories, duplicates, and whether alternatives fit the intended food group. Unknown ingredients remain unverified and are shown for review, never as confirmed safe. Relational tables let the application retrieve and update records without changing image processing. A consistent structure makes later updates easier to trace.

5.2 PROCESSING PIPELINE

- **Image Upload and OCR**

  The OCR service prepares a label photograph with grayscale conversion, denoising, contrast adjustment, thresholding, and deskewing before sending it to Tesseract.

- **Ingredient Parsing**

  OpenCV adds a light border around the processed image to reduce clipping near the edge of the label.

  The parser removes common label prefixes and percentage values, then separates items at commas, semicolons, bullets, and line breaks.

- **Profile Rule Checks**

  When an ingredient has a parenthetical group, the parser expands its listed constituents into separate terms.

  The database resolves aliases and additive codes to canonical ingredient records.

- **Results and Scan History**

Profile rules classify each match, explain the reason, and return

alternatives or a caution for unknown terms.

5.3 EVALUATION MEASURES

Evaluation should consider both the text captured from a label and the ingredient decisions made from that text. These measures help locate problems in the scan flow.

Character Error Rate

Character Error Rate (CER) compares recognized characters with a reference

transcription. It helps show how often OCR changes, omits, or adds characters.

Word Error Rate

A lower CER means the extracted label is closer to the reference text.

Word Error Rate (WER) compares recognized words with the reference list and counts substitutions, deletions, and insertions.

Allergen Precision and Recall

WER highlights errors that may change an ingredient name or separate

neighboring terms incorrectly.

Allergen precision measures how many flagged allergen matches are correct.

Allergen recall measures how many expected allergen matches the system finds.

Scan Time and Reliability

Both values matter: high precision limits unnecessary warnings, while high recall helps avoid missed matches.

Scan time and reliability can also be reviewed across image upload, OCR, parsing, database lookup, and result display. Reviewers compare clear and difficult images and note repeated failures. Reliability also depends on successful uploads, readable OCR, correct ingredient matches, and consistent explanations for the same profile. Recording image conditions makes incomplete scans easier to investigate systematically.

5.4 TEST RESULTS

The project evaluates the scan flow from image submission to the final result. Checks focus on readable and noisy label text, ingredient aliases, profile rules, and the treatment of unknown terms.

1. Profile and Dietary Choice Checks

Profile cases check that saved allergy and diet settings are loaded before a scan is evaluated.

An ingredient on the shopper's avoid list should be identified with the matching rule.

2. OCR and Ingredient Analysis

OCR and parser cases examine whether common label words and ingredient boundaries are retained.

An unknown ingredient should receive a caution for review rather than being described as confirmed safe.

3. Rule Outcomes

Alternative suggestions are checked against the status returned by the safety rules.

The result page should make the status and reason readable and leave the package available for a manual check. A failed case needs review before the planned release.

5.5 DEPLOYMENT AND USER INTERFACE

Deployment

NutriLens AI runs as a Flask web application. A shopper opens the interface in a browser, sets profile details, and uploads a label image or uses a camera where available.

The scan route sends the image to the OCR service and passes the extracted text to the parser.

A shopper can also submit ingredient text for review when a photograph is unclear.

User Interface

The interface presents SAFE, CAUTION, or AVOID together with the ingredient that triggered the rule.

Shopper Interface

Ingredient details and explanations are displayed so the shopper can

understand a result without relying on a score alone.

When records are available, the result can include alternatives and a link to earlier scans.

Administrator Interface

An administrator maintains the ingredient database, aliases, allergen

categories, explanations, and alternative records.

NutriLens AI can run locally or on a server. Its OCR and rule-based workflow does not require a dedicated graphics processor on a typical consumer computer.

CHAPTER- VI

TESTING AND VALIDATION

6.1 KNOWLEDGE BASE REVIEW

NutriLens AI does not train a predictive food-classification model. Its safety results come from ingredient records and explicit profile rules. Quality review checks canonical names, aliases, additive codes, allergen categories, explanations, and alternatives so that a label term points to the intended record. Aliases should be reviewed beside the canonical term they point to, rather than as isolated words. A spelling variation that maps to the wrong record can change the explanation shown to a shopper.

Review Steps:

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1. Canonical Ingredient Records

Ingredient entries are reviewed for spelling variants, synonyms, and consistent canonical names.

2. Alias Mapping

Allergen categories and diet-related flags are checked against the rules that use them. Test examples can check whether known aliases reach the same record and whether an unknown term stays marked for review.

3. OCR and Parser Checks

Reviewers look for duplicate or incomplete records before an update is used by the application.

4. Profile Rule Validation

Profile cases confirm that known ingredients are evaluated consistently for allergies, health needs, and avoid entries.

5. Regression Checks

Regression examples can be rerun after a data or rule change so that a known result is not altered without review. Changes to categories or explanations should be recorded so later reviewers can understand why a result changed.

6.2 TESTING PROCESS

Testing follows the NutriLens AI workflow from image upload through OCR, ingredient parsing, profile checks, and the final explanation. The aim is to confirm each service's behavior and keep a failure from producing an unsupported safety claim.

1. Unit Testing

OCR cases use readable labels and images with common quality problems.

Parser checks cover ingredient boundaries, nested groups, aliases, and additive codes.

Each function is reviewed for valid input, expected output, and clear handling of missing data.

2. Integration Testing

Profile cases check allergies, health conditions, dietary choices, and personal avoid entries.

Saved vegetarian and non-vegetarian choices should load correctly and remain mutually exclusive.

Unsupported profile controls should not appear in the interface.

3. System Testing

Integration checks follow recognized words to canonical ingredient records and then to profile rules.

Database responses are reviewed for consistent names, categories, explanations, and alternatives.

End-to-end cases follow a label image through the services to the result page.

4. Usability Testing

Scenarios include a known allergen, an avoid-list match, a safe item, and an unverified ingredient.

When text cannot be confirmed, the interface should ask the shopper to review the package.

5. Performance Review

Usability checks consider whether people can find profile settings and understand an alert.

Image review includes blur, glare, curved packaging, and uneven lighting.

Observation:

The observations help identify where scans fail and whether the result explains the reason. Test cases can be repeated after code or database updates. The project's profile tests also check that saved diet choices load as expected, that vegetarian and non-vegetarian selections cannot be active together, and that unsupported vegan controls are absent. These checks keep uncertainty visible and make changes easier to review. Repeat the same cases after changes to parsing, profile rules, or the ingredient database. Recording expected results makes differences easier to investigate before release.

6.3 VALIDATION METRICS

The measures below can be used to review OCR quality and ingredient matching.

Metric	Formula / Definition	Purpose
Character Error Rate (CER)	Character edits divided by reference characters	Measures OCR character errors
Word Error Rate (WER)	Word edits divided by reference words	Measures missing, added, or changed words
Allergen Precision and Recall	Correct matches / all flagged and expected matches	Shows how reliably relevant ingredients are found

CHAPTER- VII

CONCLUSION

7.1 CONCLUSION

NutriLens AI brings label photography, OCR, ingredient parsing, a curated database, and profile rules into one review flow. It can help shoppers identify known allergens or dietary conflicts, see the ingredient that triggered a warning, and understand the rule behind the result. When a suitable record exists, the application can also suggest an alternative. This makes the ingredient panel easier to review at the point of purchase, including when a barcode record is unavailable. The project has practical limits: a poor photograph can lead to recognition errors, and an incomplete database may not contain every ingredient or local product. Unknown terms therefore need a manual check. The current rules are deterministic and explainable, but they do not make a medical diagnosis or replace professional advice. NutriLens AI is best used as a support tool alongside the information printed on the package and guidance from a qualified clinician. This approach favors transparent rules over an unexplained health score and invites a manual check.

7.2 FUTURE SCOPE:

- **Regional Language Support**

  Add regional language and ingredient alias support so more package labels can be read and reviewed.

- **On-Device Image Recognition**

  Explore on-device image recognition to support scans when internet access is limited.

- **Nutrient Summaries**

  Add clear nutrient summaries that can be compared with a shopper’s saved profile.

- **Food Data Connections**

  Connect to trusted food-data services as coverage and project requirements grow.

REFERENCES

· *[1] R. Smith, "An Overview of the Tesseract OCR Engine," in Proc. Ninth International Conference on Document Analysis and Recognition, 2021, pp. 629–633.*

· *[2] U.S. Food and Drug Administration, "Food Allergen Labeling and Consumer Protection Act of 2004," FDA guidance document, 2022.*

· [3] European Food Safety Authority, "Guidance on the Evaluation of Food Additives and Nutrient Sources," EFSA Journal, vol. 19, no. 3, 2021.

· [4] World Health Organization, "Global Report on Diabetes and Diet-Related Noncommunicable Diseases," Geneva, 2023.

· [5] Open Food Facts Contributors, "Open Food Facts: The Free Food Products Database," project documentation, 2024.

· *[6] J. Bradski, "The OpenCV Library," Dr. Dobb's Journal of Software Tools, vol. 25, no. 11, 2000.*

· [7] S. Sicherer and H. Sampson, "Food Allergy: A Review and Update on Epidemiology and Clinical Management," Journal of Allergy and Clinical Immunology, 2018.

· [8] NutriLens AI project source code and project report, supplied with this application.

SOURCE CODE

app.py

```
from config import Config

app = Flask(__name__)

# Initialize directories

db.init_db(schema_path)

except Exception as e:

# Register blueprints

# General error handling

@app.errorhandler(404)


def page_not_found(e):


@app.errorhandler(500)

  @app.errorhandler(413)

    app = create_app()

    <html lang="en">

      <meta charset="UTF-8">

      <!-- Custom Styles -->

      <!-- Navigation Bar -->

      <div class="logo-mark">

    <!-- Guest Nav Items -->

  <!-- Auth Nav Items -->
```

```
</div>


</div>
  </nav>
    </div>
    <div class="hero-visual">
    </div>
  </div>
</div>


</div>
  </div>
  </div>
    </div>
                <div class="step-card">
      <div class="step-card">
    <div class="step-card">


    <div class="step-card">
                </div>
      </div>
    </div>


    </div>
      <div class="auth-card">
```

```
      <div class="auth-tabs">
    <div class="form-group">


    <div class="form-group">
                </form>
      <div class="form-group">
    <div class="form-group">
  <div class="form-group">
</select>

<!-- Diet Toggles -->
  </label>
<!-- Core Allergies -->


</label>
  </label>
    </label>
  </label>
</label>


</label>
</form>
</section>
</div>
```

```
</div>
<div class="panel-card">
<div class="form-group">
</label>
</label>
</button>
<!-- Camera container -->


</div>


</button>
  </div>
    </button>
    </div>
      </div>
      </li>
      </div>
    </li>
  </div>
</li>


</div>
  </div>
    </section>
    </div>
```

```
</div>
 </div>
 </div>

 </div>
 </div>

 </div>
 </div>

 <div class="panel-card">
 <!-- Pros & Cons Grid -->

 <div class="pros-card">
 <div class="cons-card">
  </div>
  </div>
  <div class="panel-card">
  </div>
  </div>
 </div>

 </div>
 </div>
```

```
        </div>
        </div>
        </section>
        </div>
        </section>
      <div class="panel-card">

      <form id="form-profile">
    <div class="form-group">
  </select>

  <!-- Diet status -->
    </label>
    </label>
    </div>
      </label>
      </label>
      </label>
      </div>
    </div>
  </div>
</form>

</section>
  <!-- Raw JSON Modal -->
```

```
    <div class="modal-body">
  </div>
</div>

<!-- Footer -->
<footer class="footer">
</footer>
    </body>
</html>
```

OCR SERVICE

```
  import cv2
import numpy as np
import pytesseract
from config import Config
@staticmethod
try:
gray = cv_image
except Exception as e:
```

```
return cv_image


@staticmethod
  try:
    # Edge detection
    # Calculate average angle
      rho, theta = line[0]
      if abs(avg_angle) > 0.5:
    center = (w // 2, h // 2)
  except Exception as e:
return cv_image


@staticmethod
  try:
  except Exception as e:
    return cv_image
    @staticmethod


    try:
    gray = cv_image


    except Exception as e:
  return cv_image
@staticmethod
```

```
try:
  # Step 1: Load image
    if img is None:
  return image_path
except:


if w < 1500 or h < 800:
cv2.THRESH_BINARY, 11, 2)
[-1, -1, -1]])
    except Exception as e:
return image_path
@staticmethod


try:
except Exception as e:
```

SAFETY RULES

```
@staticmet
  hod
  if not profile:
WHERE ua.user_id = ?
```

```
(user_id,)

(user_id,)



(user_id,)

  "allergies": allergies,

@classmethod



analyzed_it

  ems = []

# Initial status values



confidence =

  "High"

  category = "General"

# Query knowledge base
```

```
if not ing_data:
  category =
  "Unverified"
  matched_avoid = False
matched_avoid = True

if not matched_avoid:
  status =
  "CAUTION"
matched_avoid = False

matched_avoi
  d = True
  a_name = allergen['name']
status = "AVOID"
```

```
status = "CAUTION"
  if status != "AVOID":
trigger = "Vegan Diet"


if status == "SAFE":
confidence =
  "Medium"
  if status == "SAFE":
if status == "SAFE":


# Low Sugar
  Check
  confidence = "High"
# Low Sodium Check


if status == "AVOID":
overall_status = "AVOID"
```

```
analyzed_items.append({
"category": category,
  "trigger": trigger,
# LLM integration hook


if conflict_names:
  if caution_names:
  "overall_summar
  y": {
"safe_count": safe_count,


@staticmetho
  d
  import os
  if openai_key:
```

if avoids:

for item in cautions: